Effects of Verb Bias, Topicality, and Logophoric Role on Processing "zìjǐ" in Chinese

The resolution of the Chinese reflexive *zìjǐ* 'self' involves a complex integration of syntactic, pragmatic, and perspective constraints. Previous research has used syntactic Binding Theory and incorporated discourse factors like topic empathy (Huang & Liu, 2001) and logophoricity (Lyu & Kaiser, 2024) to understand the referent of *zìjǐ* 'self'. Some studies (Kaiser et al., 2009) suggest that the referent of *zìjǐ* 'self' is strongly influenced by discourse that may override syntactic constraints during comprehension. However, the pragmatic cues present in the stimuli are not always consistently controlled (e.g., Kuno & Kaburaki, 1977; Sturt, 2003), and additional confounds, such verbs' subjective or logophoric preferences (e.g., Charnavel, 2019), and the social status of sentences' subjects (e.g., *the director* vs. *secretary*) are sometimes unbalanced, potentially leading to biased referent results. Thus, we examine to what extend verb bias, topic empathy, and logophoric role may influence the antecedent resolution of *zìjǐ* 'self'.

We conducted a self-paced reading (SPR) experiment with forced-choice comprehension questions. 20 native Mandarin speakers (female=10, mean age = 21) read 128 experimental sentences. Targets all had one matrix subject (Subj1) and an embedded subject (Subj2), both shared similar levels of social impacts (controlled after norming) (Table 1). Three semantic factors: (a) <u>Logophoric role</u> was manipulated using two types of verbs to express whether the first subject (Subj1) is the **active** information **"source"** provider (e.g., with the verb *tídào* 'mention') or a **passive** information **perceiver** (e.g., with the verb *tīngshuō* 'hear'); (b) <u>verb bias</u> included verbs that tend to be ***self-directed*** (e.g., *jìchéng* 'inherit' often refers to the immediate subject of the same clause) and ***other-directed*** verbs that do not require to be associated with the immediate subject (e.g., *jīlěi* 'accumulate'); and (c) <u>topicality</u> was established by whether the Subj1 of target sentences had previously shown in the context sentence. Participants were asked to read the sentences at their own pace, and judge, between Subj1 and Subj2, which should be the referent of *zìjǐ*. Reading times (RTs) and their choice were recorded and fitted linear mixed-effects models.

Our results show that (Fig. 1) other-directed verbs strongly favored non-local binding (75%), whereas those with self-directed verbs preferred local binding (75%); topicality showed some effects but only when Subj1 was the perceiver. The inference statistic results support this observation: a strong main effect of verb bias (χ^2 (1) =3804, p<.001) and of topic empathy (p=.00038). RT data (Fig. 2) showed a significant interaction of the logophoric role and verb bias (β=.018, p =.0017) with reading faster for the source role with self-directed verbs. Additionally, topicality exerted a main effect (β= .016, p=.0056) with reading faster in non-topicality than topicality. For verb bias effects, a significant interaction between logohporic role and verb bias emerged on the aspect marker Region 12 (immediately after the verb: β= .033, p=.012). At the region "zìjǐ", verb bias was the only main effect (β=.054, p=.0019).

These findings argue against accounts that ae purely syntactic-binding-driven or simple additive models. Instead, they support a dynamic, interactive framework in which verb semantics provide an early filter, but discourse-topical information is also rapidly integrated when available and can override lexical bias. Critically, the logophoric role establishes a perspectival context that interact with topicality to modulate the resolution of referent of *zìjǐ* 'self'.

Table 1: Examples of experiment items (context sentences, followed by target sentences)

Non-topical	公司	举行	了	一	次	重要	会议。
context	company	held	PERF	one	CL	important	meeting
	'The company hold an important meeting.'						
Topical	**董事**	是	这次	重要	会议	的	发言人。
context	**director**	was	this	important	meeting	DE	spokesperson.
	'The director was the spokesperson of this important meeting.'						

Four types of target sentences

Reg 8 Subj1	**Reg 9 Logophoric role**	**Reg 10 Subj2**	**Reg 11 Verb bias**	**Reg 12**	**Reg 13 *zìjǐ***	**Reg 14**	**Reg 15**
董事 director	提到 mention 听说 hear	高管 executive	积累 accumulate 继承 inherit	了 PERF	自己 self	的 DE	资产。 assets

a. 'The director$_{Subj1}$ **mentioned** that the executive$_{Subj2}$ ***accumulated*** <u>his</u> own assets.'
b. 'The director$_{Subj1}$ **mentioned** that the executive$_{Subj2}$ ***inherit*** <u>his</u> own assets.'
c. 'The director$_{Subj1}$ **heard** that the executive$_{Subj2}$ ***accumulated*** <u>his</u> own assets.'
d. 'The director$_{Subj1}$ **heard** that the executive$_{Subj2}$ ***inherit*** <u>his</u> own assets.'

Figure 1: Distribution of respond choices across experimental conditions. Note: Error bars indicate standard errors of the proportion (SEp) (Red for local referent, blue for non-local referent)

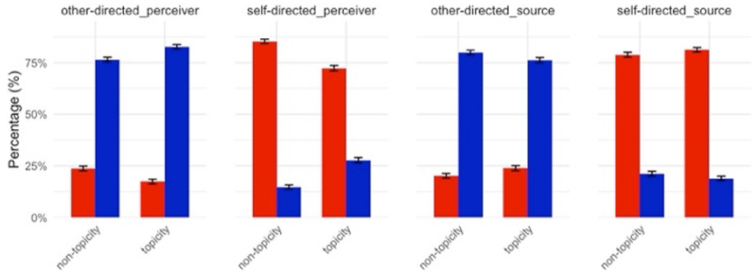
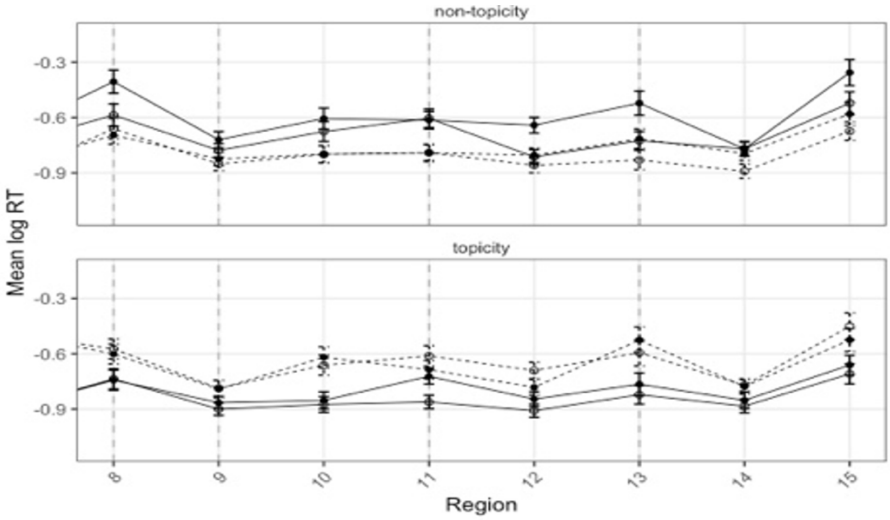
Figure 2: Log Reading Time Across Regions by Condition
(Note: solid line: perceiver, dash-line: source, dot: other-direct, circle: self-direct)